1.SYSTEM ARCHITECTURE DESIGN

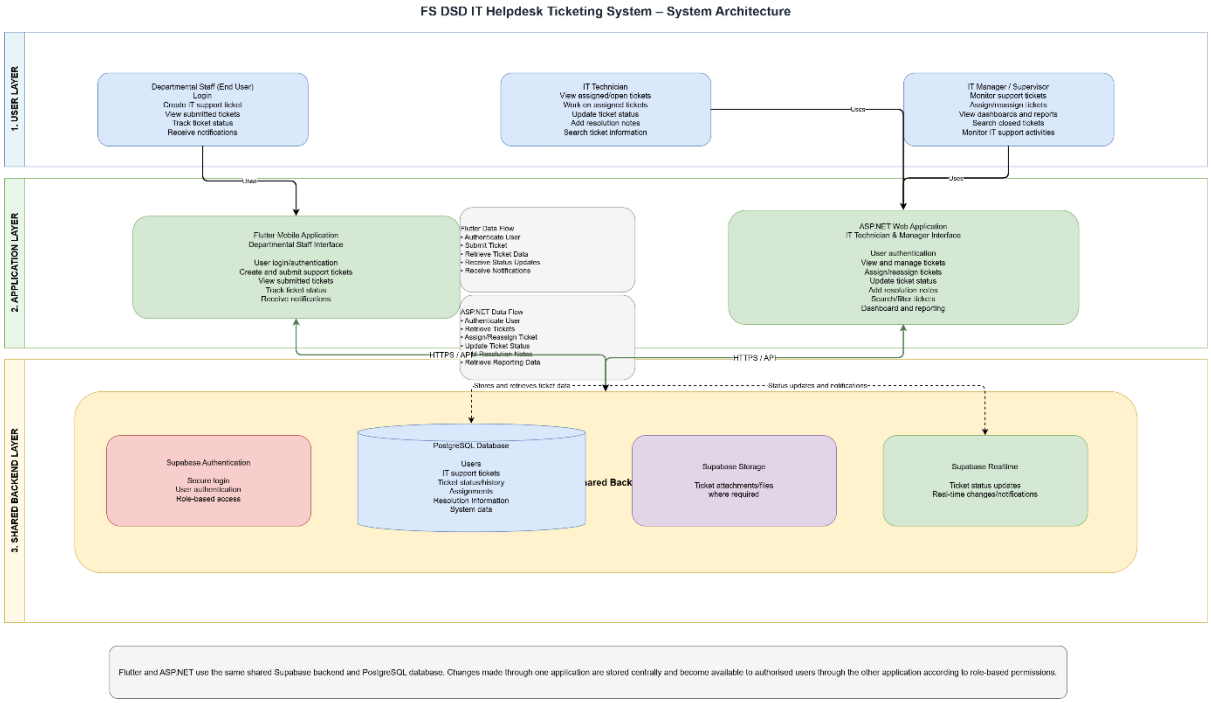


Figure 1: FS DSD IT Helpdesk Ticketing System – System Architecture

The system architecture shows how the different parts of our IT Helpdesk Ticketing System work together. The system consists mainly of a Flutter mobile application, ASP.NET web application and Supabase backend. These components share information so that departmental staff and the IT team can use the same ticketing system for different purposes.

The Flutter mobile application is mainly designed for departmental staff. Staff members will use the application to log in, report IT problems by creating support tickets, view tickets they have submitted and check the progress of their requests. They will also be able to receive updates when changes are made to their tickets.

The ASP.NET web application is mainly used by IT technicians and IT managers or supervisors. Technicians will be able to view support requests, work with tickets assigned to them, update the status of tickets and record information about how problems were resolved. Managers or supervisors will have additional functions such as assigning or reassigning tickets, monitoring support activities and viewing reports and dashboards.

Both applications use Supabase as the shared backend. Supabase Authentication is used to support secure login and control access according to the user's role. The PostgreSQL database stores important information such as users, tickets, assignments, ticket statuses, ticket history and resolution information. Supabase Storage can be used for files attached to tickets, while Realtime supports changes and updates that need to be reflected in the system.

For example, when a departmental staff member reports an IT problem using the Flutter application, the ticket information is sent to the shared backend and stored in the database. The IT team can then access that same ticket through the ASP.NET web application. When a technician updates the ticket, such as changing its status or adding resolution information, the changes are saved to the shared backend. The staff member can then see the updated information from the mobile application.

This means that the Flutter application and ASP.NET web application are not operating as separate systems. They are different interfaces for the same IT Helpdesk Ticketing System and use the same backend information. This helps keep ticket information in one place and allows each user to access the functions they need according to their role.

DESIGN SOURCE INFORMATION

Design Tool: diagrams.net (draw.io)

Design Name: FS DSD IT Helpdesk Ticketing System – System Architecture

Editable Source File: FS DSD IT HELPDESK TICKETING SYSTEM SYSTEM ARCHITECTURE.drawio

Source File Location: The original editable .drawio file will be included in the Phase 2 submission ZIP.